

NEURAL REPRESENTATIONS, NATURAL CONNECTIONS: WHAT TRANSFERS FROM HUMAN SPEECH FOUNDATION MODELS TO ANIMAL VOCALIZATIONS?

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ABSTRACT

Speech-pretrained models have shown promise in animal bioacoustics, but the factors governing their cross-species transfer remain poorly understood. We evaluate 15 frozen encoders spanning monolingual and multilingual speech, speaker verification, animal bioacoustics, and general audio on caller identification across four species and call-type classification across three. Differences between the strongest speech- and animal-pretrained representations range from -0.027 to $+0.119$ UAR; speech is significantly better in three of seven settings ($p < 0.01$) and not significantly different in the remaining four. Controlled comparisons show no systematic advantage from increased language coverage or animal-domain pretraining, while frozen speaker-verification embeddings transfer poorly. The results also show that transfer is strongly layer-dependent: raw-waveform models peak early, whereas patch-spectrogram models peak deeper.

Index Terms— Bioacoustics, Animal Vocalizations, Speech Foundation Models, Cross-Species Transfer

1. INTRODUCTION

Speech foundation models learn representations that transfer well beyond the objectives used for pretraining. Self-supervised (SSL) models such as wav2vec 2.0 [1], HuBERT [2] and WavLM [3] encode not only linguistic content but also speaker and acoustic information, with different properties emerging across layers [4]. This suggests that large-scale speech pretraining captures reusable structure in acoustic signals rather than speech-specific information alone, raising the question: *how far does this transfer extend?*

Animal vocalizations provide a particularly strong test of this question. Human speech and animal vocalizations are different communication systems, but both are acoustic signals shaped by temporal structure, spectral content, and source-specific characteristics. It is therefore natural to ask

whether representations learned from human speech remain useful across species. This is relevant in bioacoustics, where labeled datasets are often small, particularly for rare species and individual-level tasks [5].

Prior work has established that speech representations can transfer to animal vocalizations. Sarkar and Magimai-Doss [6, 7] showed that speech-pretrained SSL models encode animal caller information and later found only marginal benefits from animal-domain pretraining across several bioacoustic tasks. Cauzinille et al. [8] extended this analysis across taxa and tasks, revealing strong layer dependence and suggesting multilinguality and acoustic robustness as possible factors in transfer. Complementarily, Miron et al. [9] showed that architecture, training strategy, and data composition strongly affect bioacoustic representation transferability. Together, these studies leave open *which properties of human-speech representations and their pretraining determine successful cross-species transfer*.

Some factors that may influence such transferability are: language coverage may encourage representations less specialized to the phonetics of a single language; animal-domain pretraining may better match the target acoustics; and representations optimized for human speaker verification may preserve general patterns relevant to animal identity. In this work, we examine these factors by comparing foundation models spanning monolingual and multilingual speech, speaker verification, animal bioacoustics, and general audio across four species: rock hyrax (*Procapra capensis*), zebra finch (*Taeniopygia guttata*), little owl (*Athene noctua*), and infant marmoset (*Callithrix jacchus*). Our main contributions are:

- (1) Benchmarking 15 frozen encoders on two tasks: caller identification and call-type classification.
- (2) Testing the effects of language coverage, pretraining domain, and speaker specialization through controlled model comparisons.
- (3) Probing representations layer by layer to characterize where transferable animal-relevant information emerges.

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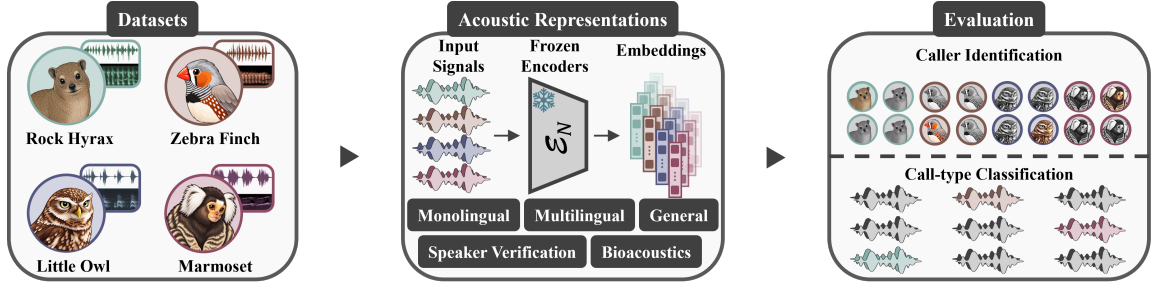


Fig. 1. Evaluation pipeline. We compare pre-trained encoders spanning monolingual (English) and multilingual speech, speaker verification, animal bioacoustics, and general audio across two tasks: caller identification and call-type classification.

2. MATERIALS & METHODS

Figure 1 summarizes the evaluation pipeline. Caller identification is evaluated on four datasets and call-type classification on three, since the little owl corpus contains only one call type in the subset considered here.

2.1. Datasets

Table 1 summarizes the four datasets.

Rock hyrax. Recordings originate from a long-term study at the Ein Gedi Nature Reserve, Israel, where 18 male hyraxes were recorded during 2002–2013 [10]. The study population was monitored at two sites within the reserve [11], and animals were individually marked and identified during field observations. Hyrax vocalizations are organized into bouts composed of three elements: *wails*, *chucks*, and *snorts*.

Zebra finch. Recordings come from a captive colony at UC Berkeley and were collected between 2011 and 2013 [12]. In total, 45 birds were recorded in a soundproof booth across 147 sessions. Caller identity was labeled during recording, and calls overlapping with another bird were excluded.

Little owl. Recordings consist of single territorial calls collected from 16 males in farmland in northern Bohemia and eastern Hungary during 2013–2014 [13]. Birds were recorded with a shotgun microphone and solid-state recorder at distances of up to 50 m, following a one-minute playback.

Infant marmoset. Recordings come from the InfantMarmosetsVox corpus, comprising five pairs of twins recorded individually in two soundproof rooms [14]. The source corpus contains approximately 58 h of annotated vocalization onset, offset, call type, and caller identity.

Pre-processing. We standardize recordings to 16 kHz mono and segment them into vocal units using their respective manual annotations. Segments shorter than the 0.4 s encoder input minimum are extended with surrounding audio for hyrax bouts and marmoset calls, and zero-padded otherwise. We also evaluated a bioacoustic denoising model to reduce differences in recording conditions. However, qualitative inspection suggested that identity-relevant frequency regions were being suppressed and was therefore not used.

Table 1. Datasets considered in this study.

	Hyrax	Finch	Owl	Marmoset
Individuals [†]	16	33	16	10
Caller-ID unit	bout	call	call	call
Call-type classes*	3	10	1	10
Caller-ID units	3,863	3,090	952	72,921
Call-type units [†]	19,970	3,071	952	72,898
Evaluated duration	89 min	21 min	20 min	464 min

*For rock hyrax, the call-type refers to vocal elements (wail, chuck, snort).

[†] Section 2.3 describes the reasons for dropping some individuals and calls.

2.2. Acoustic representations

We evaluate English speech models (wav2vec 2.0 large-lv60 [1], HuBERT-base [2], WavLM-base+ [3], and Whisper-small.en [15]), multilingual counterparts (XLS-R [16], mHuBERT [17], and XEUS [18]), speaker-verification models (wav2vec2-SV, UniSpeech-SAT-SV [19], WavLM-SV, ECAPA-TDNN [20]), animal-pretrained models (AVES-bio [21], AVES2-all, and AVES2-bio [9]), and BEATs [22] as a general-audio model. The monolingual models wav2vec 2.0, HuBERT, and WavLM are SSL with a raw-waveform front end, while Whisper is trained for ASR with a log-mel spectrogram (+1D convolutions) front end. XLS-R-300m and mHuBERT-147 are architecture-matched to wav2vec 2.0 and HuBERT, respectively. For speaker verification, all models except ECAPA-TDNN combine an SSL encoder with an x-vector head; ECAPA-TDNN provides a single speaker embedding. WavLM-SV’s encoder matches WavLM-base+. AVES-bio has a HuBERT-based encoder pretrained on bioacoustic audio, while the BEATs/AVES2-all/AVES2-bio series enables a controlled comparison of pretraining domain with the backbone fixed. All embeddings are pooled by mean and standard deviation, resulting in a 2048-D feature vector for XEUS, XLS-R, and wav2vec 2.0 large, 384-D for ECAPA, and 1538-D for the rest.

Baseline. As a non-pre-trained acoustic baseline, we use 20 Mel-frequency cepstral coefficients (MFCCs) with first- and second-order derivatives from 25 ms windows every 10 ms, mean-and-standard-deviation pooled and concatenated with log-duration, resulting in a 121-D feature vector.

2.3. Evaluation

We perform classification using 3-fold cross-validation.

Data splits. For caller identification, folds are grouped by recording session. Individuals with < 3 sessions have their largest session split into contiguous blocks so that each individual appears in all folds. This affects 2/16 hyraxes, 13/33 zebra finches, and all 16 little owls, whose corpus has only two recording partitions; no marmoset is affected (Section 3). For call-type classification, folds are grouped by caller, so evaluation is performed on unseen individuals. In both tasks, every target class must contain at least five samples per fold to avoid unstable per-class recall estimates.

Classification. Embeddings are standardized and classified with a logistic regression using balanced class weights. Layer and C are selected jointly by 2-fold cross-validation within each training fold, then refit on the full training split and evaluated once on the held-out fold. Performance is reported as mean UAR across folds.

3. EXPERIMENTS & RESULTS

Tables 2 and 3 summarize the results for caller identification and call-type classification, respectively. For each encoder, we report the UAR at the layer selected on the inner folds of each split (Section 2.3). MFCCs provide a reference under the same evaluation protocol. Pairwise differences are assessed by a paired cluster bootstrap that resamples recording sessions (caller identification) or callers (call-type classification) within each fold. All p -values reported below are Holm–Bonferroni adjusted within each comparison family.

Table 2. Caller identification UARs; bold marks the best encoder per corpus.

Domain	Acoustic representation	Hyrax n=16	Finch n=33	Owl n=16	Marmoset n=10
Baseline	MFCC	0.328	0.455	0.649	0.261
Monolingual	HuBERT-base	0.373	0.480	0.710	0.403
	wav2vec2-large	0.408	0.530	0.725	0.421
	WavLM-base+	0.392	0.497	0.716	0.391
	Whisper-small.en	0.456	0.581	0.708	0.482
Multilingual	mHuBERT-147	0.384	0.521	0.700	0.389
	XEUS	0.342	0.205	0.656	0.326
	XLS-R-300m	0.400	0.536	0.729	0.414
Speaker verification	ECAPA-TDNN	0.253	0.375	0.508	0.260
	UniSpeech	0.372	0.464	0.695	0.412
	wav2vec2-SV	0.374	0.443	0.681	0.339
	WavLM-SV	0.392	0.497	0.716	0.391
Animal	AVES-bio	0.421	0.485	0.725	0.389
	AVES2-all	0.445	0.459	0.734	0.409
	AVES2-bio	0.474	0.468	0.745	0.415
General	BEATs	0.456	0.469	0.757	0.433

Cross-species transfer. We first compare the best speech-pretrained and animal-pretrained representations in each task–dataset setting, both selected on the inner folds. Speech

Table 3. Call-type classification UARs; bold marks the best encoder per corpus.

Domain	Acoustic representation	Hyrax n=3	Finch n=10	Marmoset n=10
Baseline	MFCC	0.873	0.780	0.579
Monolingual	HuBERT-base	0.911	0.822	0.699
	wav2vec2-large	0.929	0.828	0.664
	WavLM-base+	0.931	0.799	0.689
	Whisper-small.en	0.913	0.866	0.717
Multilingual	mHuBERT-147	0.932	0.827	0.698
	XEUS	0.635	0.501	0.515
	XLS-R-300m	0.939	0.827	0.654
Speaker verification	ECAPA-TDNN	0.865	0.729	0.469
	UniSpeech	0.918	0.810	0.708
	wav2vec2-SV	0.908	0.818	0.655
	WavLM-SV	0.931	0.799	0.689
Animal	AVES-bio	0.923	0.802	0.681
	AVES2-all	0.860	0.771	0.731
	AVES2-bio	0.858	0.795	0.726
General	BEATs	0.882	0.802	0.717

models are significantly better in three of the seven settings: on zebra finch for caller identification ($+0.119$ UAR, $p < 0.001$) and call-type classification ($+0.036$, $p = 0.003$), and on marmoset for caller identification ($+0.086$, $p < 0.001$). Whisper is the selected speech encoder in all three settings, in every fold but one. The remaining four settings show no significant difference: hyrax caller identification (-0.019 , $p = 0.17$), little-owl caller identification (-0.027 , $p = 0.21$), hyrax call-type classification ($+0.006$, $p = 0.20$), and marmoset call-type classification (-0.014 , $p = 0.07$). In no setting is the animal encoder significantly better.

Language coverage. Across the 14 architecture-matched HuBERT–mHuBERT and wav2vec 2.0–XLS-R comparisons, absolute differences never exceed 0.040 UAR. Three are significant, all favoring the multilingual model: mHuBERT on zebra finch caller identification ($+0.040$, $p < 0.001$) and hyrax call-type classification ($+0.021$, $p = 0.002$), and XLS-R on hyrax call-type classification ($+0.010$, $p < 0.001$). The remaining eleven, including all four marmoset and both little-owl comparisons, are not significant. Thus, increased language coverage provides at most a small and inconsistent advantage for cross-species transfer.

Pretraining domain. With the BEATs backbone fixed, differences among BEATs, AVES2-all, and AVES2-bio are at most 0.029 UAR for caller identification and 0.031 for call-type classification. Of 14 comparisons against BEATs, five are significant: four favor BEATs (hyrax call type and marmoset caller identification, both variants, $p \leq 0.017$) and one favors an animal-pretrained variant (AVES2-all on marmoset call type, $+0.014$, $p < 0.001$). Animal-domain pretraining therefore provides no consistent advantage.

Speaker specialization. Speaker-verification models transfer like other raw-waveform speech models. WavLM-SV’s encoder matches WavLM-base+. Their x-vector heads, in contrast, score below every encoder layer of the same

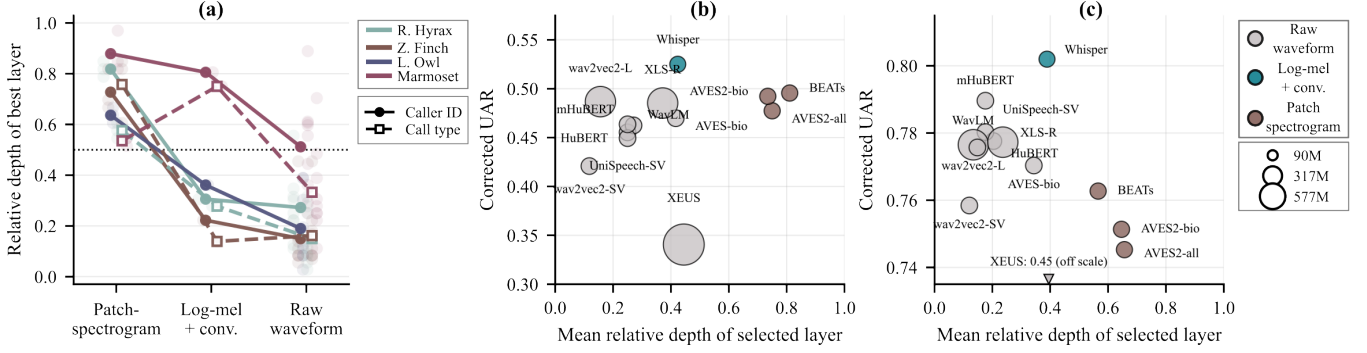


Fig. 2. (a) Mean selected relative layer depth by front end and task. (b,c) Mean selected depth versus chance-corrected UAR for caller ID and call-type classification. Bubble area denotes model size; color denotes front end. XEUS is off-scale in (c).

model in all 21 settings, by 0.10–0.32 UAR ($p < 0.001$ in all cases), and below the MFCC baseline in 26 of 28 settings. ECAPA-TDNN, which provides only one embedding, is below MFCC in all seven. Speaker-verification layers therefore transfer poorly, while the layers beneath them do not; however, adapting the encoder could yield different behavior.

Session independence. Figure 3 shows the results when the caller identification experiments are restricted to session-independent individuals. Hyrax is essentially unchanged. For zebra finch, the corpus UAR of 0.581 decomposes into 0.81 for the 13 block-split birds and 0.44 for the 20 session-independent ones. Little owl drops by 0.19–0.37 for every encoder. Call-type folds hold out callers and are unaffected.

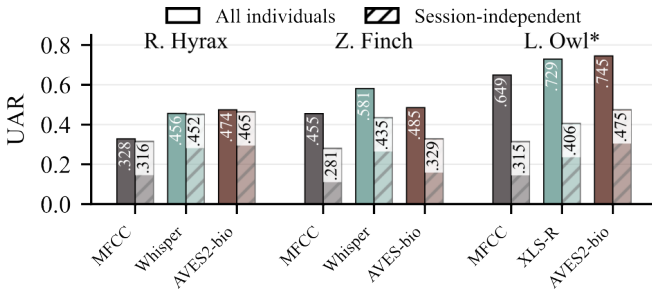


Fig. 3. Caller identification UAR for all individuals (solid) and for session-independent individuals only. *Little owl: two-fold across its two recording partitions.

Layer-wise transfer. A pattern we found across encoders is that selected layer depth is more strongly associated with the front-end representation than with the pretraining domain (Figure 2a). For caller identification, raw-waveform models (e.g. wav2vec) select earlier layers (0.15–0.51 of relative network depth, averaged across folds), whereas patch-spectrogram BEATs/AVES2 models select deeper ones (0.64–0.88). Animal-pretrained AVES-bio follows the raw-waveform speech models, while BEATs and AVES2, which share the same front end, select similar depths. Whisper falls between these groups on caller identification and varies across

datasets. Call-type classification shows the same pattern on hyrax and zebra finch, but less clearly on marmoset.

Figures 2b and 2c show each encoder by its mean depth and mean chance-corrected UAR i.e., $(\text{UAR} - \text{chance}) / (1 - \text{chance})$. Whisper is the strongest encoder in both tasks at intermediate depth. For call-type classification, all raw-waveform speech encoders cluster in the first quarter of the network within 0.03 corrected UAR of each other, while the patch-spectrogram encoders sit deeper and slightly lower. Model size shows no relation to transfer: XEUS is the largest encoder and the weakest in both tasks. Since the front end is confounded with model family, this should be interpreted as an association rather than a controlled comparison. Nevertheless, the results show that evaluating only the final layer can mischaracterize cross-species transfer.

4. CONCLUSION

Human-speech representations transfer across species for both caller identity and call type: the strongest speech models significantly outperform animal-pretrained models in three of seven settings and do not differ significantly in the remaining four. For hyrax and marmoset, this holds across recording sessions, whereas zebra finch and little-owl caller identification partly reflects within-session recognition.

Controlled comparisons show no systematic advantage from increased language coverage or animal-domain pre-training. Speaker-verification embeddings transfer poorly to animal identity, although useful information remains in their underlying encoders. Transfer is also strongly layer-dependent, with the most informative representation occurring at different depths across front ends.

Because all encoders are evaluated with frozen weights, these results characterize information directly accessible from pretrained representations, not the full potential of their architectures. Supervised adaptation may therefore change these conclusions, as prior work shows that post-training SSL encoders can substantially improve bioacoustic transfer [9].

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Compliance with Ethical Standards: We performed no new animal procedures for this work. All recordings used were collected in earlier studies under the ethical approvals reported therein (see section 2.1) and were obtained for the present analysis under the terms under which their authors made them available.

5. REFERENCES

- [1] Alexei Baevski, Yuhao Zhou, Abdelrahman Mohamed, and Michael Auli, “wav2vec 2.0: A framework for self-supervised learning of speech representations,” *In Proc. NeurIPS*, vol. 33, pp. 12449–12460, 2020.
- [2] Wei-Ning Hsu, Benjamin Bolte, et al., “HuBERT: Self-supervised speech representation learning by masked prediction of hidden units,” *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, vol. 29, pp. 3451–3460, 2021.
- [3] Sanyuan Chen, Chengyi Wang, Zhengyang Chen, et al., “WavLM: Large-scale self-supervised pre-training for full stack speech processing,” *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1505–1518, 2022.
- [4] Ankita Pasad, Ju-Chieh Chou, and Karen Livescu, “Layer-wise analysis of a self-supervised speech representation model,” in *Proc. ASRU*, 2021, pp. 914–921.
- [5] Elly Knight et al., “Individual identification in acoustic recordings,” *Trends in Ecology & Evolution*, vol. 39, no. 10, pp. 947–960, 2024.
- [6] Eklavya Sarkar and Mathew Magimai-Doss, “Can self-supervised neural representations pre-trained on human speech distinguish animal callers?,” in *Proc. INTERSPEECH*, 2023, pp. 1189–1193.
- [7] Eklavya Sarkar and Mathew Magimai-Doss, “Comparing self-supervised learning models pre-trained on human speech and animal vocalizations for bioacoustics processing,” in *Proc. ICASSP*, 2025, pp. 1–5.
- [8] Jules Cauzinille et al., “Crossing the species divide: Transfer learning from speech to animal sounds,” in *Proc. DCASE*, 2025, pp. 50–54.
- [9] Marius Miron, David Robinson, Milad Alizadeh, et al., “What matters for bioacoustic encoding,” in *Proc. ICLR*, 2026.
- [10] Vlad Demartsev et al., “Male rock hyraxes that maintain an isochronous song rhythm achieve higher reproductive success,” *Journal of Animal Ecology*, vol. 92, no. 8, pp. 1520–1531, 2023.
- [11] Lee Koren and Eli Geffen, “Complex call in male rock hyrax (*Procavia capensis*): A multi-information distributing channel,” *Behavioral Ecology and Sociobiology*, vol. 63, no. 4, pp. 581–590, 2009.
- [12] Julie E. Elie and Frédéric E. Theunissen, “Zebra finches identify individuals using vocal signatures unique to each call type,” *Nature Communications*, vol. 9, pp. 4026, 2018.
- [13] Dan Stowell et al., “Automatic acoustic identification of individuals in multiple species: improving identification across recording conditions,” *Journal of The Royal Society Interface*, vol. 16, no. 153, pp. 20180940, 2019.
- [14] Ya-Jie Zhang et al., “Automatic detection and classification of marmoset vocalizations using deep and recurrent neural networks,” *The Journal of the Acoustical Society of America*, vol. 144, no. 1, pp. 478–487, 2018.
- [15] Alec Radford et al., “Robust speech recognition via large-scale weak supervision,” in *Proc. ICML*, 2023, pp. 28492–28518.
- [16] Arun Babu et al., “XLS-R: Self-supervised cross-lingual speech representation learning at scale,” in *Proc. INTERSPEECH*, 2022, pp. 2278–2282.
- [17] Marceley Zanon Boito, Vivek Iyer, Nikolaos Lagos, Laurent Besacier, and Ioan Calapodescu, “mHuBERT-147: A compact multilingual HuBERT model,” in *Proc. INTERSPEECH*, 2024, pp. 3939–3943.
- [18] William Chen, Wangyou Zhang, Yifan Peng, Xinjian Li, Jinchuan Tian, et al., “Towards robust speech representation learning for thousands of languages,” in *Proc. EMNLP*, 2024, pp. 10205–10224.
- [19] Sanyuan Chen et al., “UniSpeech-SAT: Universal speech representation learning with speaker aware pre-training,” in *Proc. ICASSP*, 2022, pp. 6152–6156.
- [20] Brecht Desplanques et al., “ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification,” in *Proc. INTERSPEECH*, 2020, pp. 3830–3834.
- [21] Masato Hagiwara, “AVES: Animal vocalization encoder based on self-supervision,” in *Proc. ICASSP*, 2023.
- [22] Sanyuan Chen, Yu Wu, Chengyi Wang, et al., “BEATs: Audio pre-training with acoustic tokenizers,” in *Proc. ICML*, 2023, pp. 5178–5193.